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## KCS

- KCS stands for Knowledge-Centered Service or Knowledge-Centered Support .
- It is a best practice methodology that provides a detailed description of how service organizations can work more effectively with knowledge.
- The main goal of KCS is to improve service delivery, become more productive, decrease costs, and increase service levels to customers.
- KCS is based on four basic principles :
  - Create content as a by-product of solving problems.
  - Evolve content based on demand and usage.
  - Develop a knowledge base of collective experience.
  - Reward learning, collaboration, sharing, and improvement.
- KCS has two main processes: the Solve Loop and the Evolve Loop .
  - The Solve Loop is the process of capturing, structuring, and reusing knowledge during the problem-solving process.
  - The Evolve Loop is the process of analyzing, improving, and governing the knowledge base and the KCS practices.
- KCS has several benefits for service organizations, such as :
  - Increased customer satisfaction and loyalty.
  - Reduced time to resolution and repeat incidents.
  - Improved operational efficiency and scalability.
  - Enhanced employee engagement and retention.
  - Increased innovation and value creation.

I hope this information is helpful for you. If you have any questions or feedback, please let me know.

I am Sydney, your AI assistant. I will help you with your topic. Here is the content in markdown format:

### **Unit 1 - Brief history of microbiology, Types of microorganisms, Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria, lichens, algae, protozoa, helminthes, viral structures, viral multiplication, Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

#### **Brief history of microbiology**

- Microbiology is the study of microscopic organisms, such as bacteria, viruses, fungi, and protozoa.

- Microbiology essentially began with the development of the microscope in the 17th century.
- Antonie van Leeuwenhoek (1632–1723) was the first to observe and document microorganisms using a microscope of his own design .
- Robert Hooke (1635–1703) was the first to use a microscope to observe living things, such as fungi and cells .
- Ferdinand Cohn (1828–1898) was the founder of bacteriology, a subdiscipline of microbiology, and described several bacteria, such as *Bacillus* and *Beggiatoa*.
- Louis Pasteur (1822–1895) was a pioneer of microbiology and demonstrated the role of microorganisms in fermentation, spoilage, and disease.
- Robert Koch (1843–1910) was a microbiologist who established the germ theory of disease and developed methods to isolate and identify pathogens.
- Other notable microbiologists include Alexander Fleming (1881–1955), who discovered penicillin, the first antibiotic; Martinus Beijerinck (1851–1931), who coined the term virus and discovered the first virus; and Carl Woese (1928–2012), who proposed the three-domain system of classification based on molecular phylogeny.

### **Types of microorganisms**

- Microorganisms are classified into different groups based on their structure, function, and evolutionary relationships.
- The three-domain system of classification divides all living organisms into three domains: Bacteria, Archaea, and Eukarya.
- Bacteria and Archaea are prokaryotes, which means they lack a nucleus and other membrane-bound organelles. They are usually unicellular and have diverse shapes, metabolisms, and habitats.
- Eukarya are eukaryotes, which means they have a nucleus and other membrane-bound organelles. They can be unicellular or multicellular and include animals, plants, fungi, and protists.
- Protists are a diverse group of eukaryotic microorganisms that do not fit into the other kingdoms. They include algae, protozoa, slime molds, and water molds.
- Fungi are eukaryotic microorganisms that are heterotrophic, which means they obtain organic molecules from other sources. They include yeasts, molds, and mushrooms.
- Viruses are non-living entities that consist of genetic material (DNA or RNA) enclosed in a protein coat. They can only replicate inside a host cell and cause various diseases.

### **Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria**

- Bacteria are the most abundant and diverse group of prokaryotes. They can be found in almost every environment and play important roles in the

biosphere.

- Bacteria are classified into different phyla based on their genetic and phenotypic characteristics. One of the major criteria for classification is the Gram stain, which differentiates bacteria based on the structure of their cell wall.
- Gram-positive bacteria have a thick layer of peptidoglycan, a polymer of sugar and amino acids, in their cell wall. They stain purple with the Gram stain and are resistant to some antibiotics, such as penicillin.
- Gram-negative bacteria have a thin layer of peptidoglycan and an outer membrane composed of lipopolysaccharides, lipoproteins, and phospholipids, in their cell wall. They stain pink with the Gram stain and are more resistant to antibiotics, such as lysozyme.
- Proteobacteria are a large and diverse phylum of Gram-negative bacteria that include many pathogens, such as *Escherichia coli*, *Salmonella*, *Vib*

### ### Brief history of microbiology

- Microbiology is the scientific study of microorganisms, which are usually too small to be
- Microbiology has its origins in the 16th century, when some people observed the existence
- The development of the microscope in the 17th century enabled the direct observation of m
- Robert Hooke (1635-1703) was an English scientist who coined the term "cell" after observi
- Anton van Leeuwenhoek (1632-1723) was a Dutch draper and amateur microscopist who made his
- In the 18th and 19th centuries, microbiology progressed with the discoveries of spontaneou
- Spontaneous generation was the idea that living things could arise from non-living matter.
- Louis Pasteur (1822-1895) was a French chemist and microbiologist who made many contributi
- Germ theory of disease was the hypothesis that microorganisms were the cause of infectious
- Robert Koch (1843-1910) was a German physician and microbiologist who established the criti
- Joseph Lister (1827-1912) was a British surgeon and pioneer of antiseptic surgery, who use
- Ignaz Semmelweis (1818-1865) was a Hungarian physician and obstetrician who discovered tha
- Microbial taxonomy is the classification and naming of microorganisms based on their simil
- Robert Whittaker (1920-1980) was an American botanist and ecologist who proposed a five-ki
- Carl Woese (1928-2012) was an American microbiologist and biophysicist who proposed a thre
- In the 21st century, microbiology continues to advance with the development of new technic

## Types of microorganisms

Microorganisms are microscopic organisms that exist as unicellular, multicellular, or cell clusters. They can be beneficial or harmful to life, and they can be divided into six major types :

- **Bacteria:** These are prokaryotic cells, meaning they lack a nucleus and other membrane-bound organelles. They have a cell wall made of peptidoglycan, and they can be classified by their shape, Gram stain, metabolism, and genetic characteristics. Some examples of bacteria are *Escherichia coli*, *Staphylococcus aureus*, and *Streptococcus pyogenes*.
- **Archaea:** These are also prokaryotic cells, but they differ from bacteria

in their cell wall composition, membrane structure, and gene expression. They are often found in extreme environments, such as hot springs, salt lakes, and deep-sea vents. Some examples of archaea are *Methanobacterium*, *Halobacterium*, and *Thermococcus*.

- **Protozoa:** These are eukaryotic cells, meaning they have a nucleus and other membrane-bound organelles. They are usually motile and heterotrophic, meaning they obtain their nutrients by ingesting other organisms or organic matter. They can be classified by their mode of locomotion, such as flagella, cilia, or pseudopodia. Some examples of protozoa are *Paramecium*, *Amoeba*, and *Plasmodium*.
- **Algae:** These are also eukaryotic cells, but they are photosynthetic, meaning they use light energy to produce organic molecules. They have a cell wall made of cellulose, and they can be unicellular or multicellular. They can be classified by their pigments, such as chlorophyll, carotenoids, or phycobilins. Some examples of algae are *Spirogyra*, *Chlamydomonas*, and *Ulva*.
- **Fungi:** These are also eukaryotic cells, but they are saprophytic, meaning they decompose organic matter. They have a cell wall made of chitin, and they can be unicellular or multicellular. They can be classified by their morphology, such as yeasts, molds, or mushrooms. Some examples of fungi are *Saccharomyces*, *Aspergillus*, and *Agaricus*.
- **Viruses:** These are acellular entities, meaning they are not composed of cells. They consist of a nucleic acid (DNA or RNA) enclosed in a protein coat, and sometimes a lipid envelope. They can only replicate inside a host cell, and they can infect all types of organisms. They can be classified by their shape, size, genome, and host range. Some examples of viruses are Influenza, HIV, and Herpes.

### Basic idea of domain bacteria

- Domain bacteria is one of the three domains of life, along with domain archaea and domain eukarya.
- Domain bacteria includes the prokaryotes people encounter on an everyday basis. Prokaryotes are organisms that lack a membrane-bound nucleus and other organelles.
- Most bacterial species are heterotrophic; that is, they acquire their food from organic matter. The largest number of bacteria are saprobic, meaning that they feed on dead or decaying organic matter. A few bacterial species are parasitic; these bacteria live within host organisms and cause disease.
- Bacteria can also be classified by their shape, cell wall structure, metabolic pathways, and response to oxygen.
- The domain bacteria is divided into several phyla, such as proteobacteria, cyanobacteria, firmicutes, actinobacteria, and spirochaetes. Each phylum has its own characteristics and diversity of species.
- Bacteria play important roles in the production of industrial chemicals and pharmaceuticals, such as antibiotics, enzymes, vitamins, hormones,

and biofuels . Bacteria also participate in various biogeochemical cycles, such as nitrogen, carbon, and sulfur cycles .

- Bacteria can be studied using various methods, such as microscopy, staining, culturing, molecular biology, and bioinformatics . Bacteria can also be genetically modified for various purposes, such as bioremediation, biotechnology, and medicine .

### **Proteobacteria**

- Proteobacteria is a phylum of gram-negative bacteria discovered by Carl Woese in the 1980s based on nucleotide sequence homology .
- Proteobacteria are further classified into the classes alpha-, beta-, gamma-, delta- and epsilonproteobacteria, each class having separate orders, families, genera, and species.
- Proteobacteria include a wide variety of pathogenic and free-living bacteria, such as *Escherichia*, *Salmonella*, *Vibrio*, *Yersinia*, *Legionella*, and many others.
- Proteobacteria are also involved in nitrogen fixation, carbon monoxide oxidation, and sulfur cycling .
- Proteobacteria have diverse shapes, metabolisms, and habitats, ranging from soil and water to animal and plant hosts .

Some examples of proteobacteria are:

- Alphaproteobacteria: oligotrophs, organisms capable of living in low-nutrient environments such as deep oceanic sediments, glacial ice, or deep undersurface soil. Some examples are *Rhizobium* (nitrogen-fixing symbionts of legumes), *Rickettsia* (intracellular parasites of animals), and *Caulobacter* (stalked bacteria that attach to surfaces).
- Betaproteobacteria: mostly aerobic or facultative anaerobic bacteria that can use various organic compounds as electron donors or acceptors. Some examples are *Neisseria* (pathogens of the human mucous membranes), *Nitrosomonas* (nitrifying bacteria that oxidize ammonia to nitrite), and *Burkholderia* (opportunistic pathogens and bioremediators).
- Gammaproteobacteria: the largest and most diverse class of proteobacteria, comprising bacteria that can use a variety of substrates and energy sources. Some examples are *Pseudomonas* (versatile and ubiquitous bacteria that can degrade various pollutants), *Escherichia* (intestinal bacteria that can cause diarrhea and urinary tract infections), and *Vibrio* (marine bacteria that can cause cholera and food poisoning).
- Deltaproteobacteria: mostly anaerobic bacteria that are involved in sulfur and iron cycling. Some examples are *Desulfovibrio* (sulfate-reducing bacteria that can corrode metal pipes), *Bdellovibrio* (predatory bacteria that invade and consume other gram-negative bacteria), and *Myxococcus* (social bacteria that form fruiting bodies when starved).
- Epsilonproteobacteria: mostly microaerophilic or anaerobic bacteria that inhabit extreme environments such as hot springs, hydrothermal vents,

and animal guts. Some examples are *Helicobacter* (gastric pathogens that can cause ulcers and cancer), *Campylobacter* (intestinal pathogens that can cause diarrhea and dysentery), and *Wolinella* (hydrogen-oxidizing bacteria that live in the rumen of cows).

### **Nonproteobacteria Gram-Negative Bacteria**

- Nonproteobacteria are a diverse group of bacteria that do not belong to the phylum Proteobacteria, which contains the majority of gram-negative bacteria.
- Nonproteobacteria can be either gram-negative or gram-positive, depending on the presence or absence of an outer membrane in their cell wall.
- Nonproteobacteria can have various metabolic modes, such as photoautotrophy, chemoheterotrophy, or chemoautotrophy.
- Nonproteobacteria can be either motile or nonmotile, and some have distinctive shapes, such as spiral or stalked.
- Some examples of nonproteobacteria are:
  - Chlamydia: a group of obligate intracellular parasites that cause various diseases in humans and animals, such as chlamydia, trachoma, and psittacosis. They have a unique developmental cycle that involves two forms: elementary bodies and reticulate bodies.
  - Spirochetes: a group of motile, spiral-shaped bacteria that have a flexible cell wall and a unique flagellar arrangement called axial filaments. They can cause diseases such as syphilis, Lyme disease, and leptospirosis. They are difficult or impossible to culture in the laboratory.
  - CFB group: a phylum of gram-negative nonproteobacteria that includes the genera *Cytophaga*, *Fusobacterium*, and *Bacteroides*. They share some similarities in their DNA sequences, but have diverse morphologies and metabolisms. *Cytophaga* are rod-shaped bacteria that degrade cellulose and other polysaccharides. *Fusobacterium* are spindle-shaped bacteria that are involved in periodontal disease and other infections. *Bacteroides* are anaerobic bacteria that are abundant in the human gut and play a role in digestion and immunity.
  - Planctomycetes: a group of gram-negative nonproteobacteria that have a unique cell structure and life cycle. They have a membrane-bound compartment called the pirellulosome that contains the nucleoid and ribosomes. They also have a stalk-like appendage called the holdfast that attaches them to surfaces. They can reproduce by budding or binary fission.
  - Phototrophic bacteria: a group of bacteria that use light as an energy source and can perform photosynthesis. They can be either gram-

negative or gram-positive, and can have different types of pigments, such as chlorophyll, bacteriochlorophyll, or carotenoids. They can be classified into two major groups: oxygenic and anoxygenic. Oxygenic phototrophic bacteria produce oxygen as a byproduct of photosynthesis, such as cyanobacteria. Anoxygenic phototrophic bacteria do not produce oxygen, and use other electron donors, such as hydrogen sulfide or organic compounds, such as purple sulfur bacteria and green sulfur bacteria.

### Gram-positive bacteria

- Gram-positive bacteria are bacteria that have a thick cell wall composed of peptidoglycan and teichoic acids .
- Gram-positive bacteria can be distinguished from gram-negative bacteria by the Gram stain test, which involves a chemical dye that stains the cell wall purple or blue .
- Gram-positive bacteria include many important pathogens, such as Staphylococcus, Streptococcus, Bacillus, Clostridium, Corynebacterium, Listeria, and Mycobacterium .
- Gram-positive bacteria can cause various infections, such as skin infections, pneumonia, meningitis, anthrax, tetanus, diphtheria, tuberculosis, and leprosy .
- Gram-positive bacteria can also have beneficial roles, such as producing antibiotics, fermented foods, probiotics, and biodegradable plastics.
- Gram-positive bacteria can be treated with antibiotics that target their cell wall synthesis, such as penicillins, cephalosporins, vancomycin, and daptomycin .
- Gram-positive bacteria can develop resistance to antibiotics by altering their cell wall structure, producing enzymes that inactivate the drugs, or acquiring genes that confer resistance .

### Lichens

- Lichens are **composite organisms** that consist of a **symbiotic association** of algae (usually green) or cyanobacteria and fungi (mostly ascomycetes and basidiomycetes).
- The dominant partner is the **fungus**, which gives the lichen the majority of its characteristics, from its **thallus shape** to its **fruiting bodies**.
- The alga or cyanobacterium provides **photosynthesis** and **nitrogen fixation** for the fungus, while the fungus provides **protection**, **water**, and **minerals** for the alga or cyanobacterium.
- Lichens can be classified into four main types based on their thallus shape: **crustose** (crusty or flat), **foliose** (leafy or lobed), **fruticose** (shrubby or branched), and **squamulose** (scaly or granular).
- Lichens are found **worldwide** and occur in a variety of environmental conditions, from **arctic tundra** to **tropical rainforest**.

- Lichens are **pioneer species**, among the first living things to grow on bare rock or areas denuded of life by a disaster.
- Lichens have many **ecological roles**, such as **bioindicators** of air quality, **decomposers** of organic matter, **soil stabilizers** and **erosion preventers**, and **food sources** and **nesting materials** for animals .
- Lichens also have many **human uses**, such as **dyes**, **perfumes**, **medicines**, **litmus paper**, and **lichenometry** (a method of dating surfaces based on lichen growth).

## Algae

- Algae are a large and diverse group of photosynthetic eukaryotic organisms that include seaweeds and many single-celled forms .
- Algae are not plants, but they share some characteristics with them, such as having chlorophyll, cell walls, and the ability to produce oxygen .
- Algae belong to the kingdom Protista, which is a polyphyletic group of organisms that do not fit into any other kingdom .
- Algae can be classified into different groups based on their pigments, cell structure, storage products, and life cycles .
- Some of the major groups of algae are:
  - Green algae (Chlorophyta): They have chlorophyll a and b, store starch, and have cellulose in their cell walls. They are the most closely related to plants. Some examples are Spirogyra, Volvox, and Ulva .
  - Brown algae (Phaeophyta): They have chlorophyll a and c, and a brown pigment called fucoxanthin. They store laminarin and mannitol, and have alginic acid in their cell walls. They are mostly marine and include the largest algae, such as kelps .
  - Red algae (Rhodophyta): They have chlorophyll a and d, and a red pigment called phycoerythrin. They store floridean starch, and have agar and carrageenan in their cell walls. They are mostly marine and can live in deep waters .
  - Diatoms (Bacillariophyta): They have chlorophyll a and c, and a yellow-brown pigment called diatoxanthin. They store oil or chrysolaminarin, and have silica in their cell walls. They are mostly aquatic and have a variety of shapes and patterns .
  - Dinoflagellates (Dinophyta): They have chlorophyll a and c, and a red or brown pigment called peridinin. They store starch or oil, and have cellulose plates in their cell walls. They are mostly aquatic and have two flagella. Some of them can produce bioluminescence or toxins .
  - Euglenoids (Euglenophyta): They have chlorophyll a and b, and a yellow-green pigment called xanthophyll. They store paramylon, and have no cell wall but a proteinaceous pellicle. They are mostly freshwater and have one or two flagella. Some of them can lose their



chloroplasts and become heterotrophic .

- Algae have many types of life cycles, ranging from simple asexual reproduction to complex alternation of generations .
- Algae play an important role in the production of industrial chemicals and pharmaceuticals, such as agar, alginates, carrageenans, iodine, biofuels, antioxidants, and anti-inflammatory agents .

## Protozoa

- Protozoa are **microscopic unicellular eukaryotes** that have a relatively complex internal structure and carry out complex metabolic activities .
- Some protozoa have structures for **propulsion or other types of movement**, such as flagella, cilia, or pseudopodia .
- Protozoa can be either **free-living or parasitic**, that feed on organic matter such as other microorganisms or organic tissues and debris .
- Protozoa are **heterotrophic**, meaning they use organic carbon as a source of energy .
- Protozoa have a **membrane-bound nucleus** and other organelles, such as mitochondria, endoplasmic reticulum, Golgi apparatus, and lysosomes .
- Protozoa can reproduce **asexually** by binary fission, multiple fission, budding, or schizogony, or **sexually** by conjugation, syngamy, or gametogony .
- Protozoa are currently classified into **six phyla** on the basis of light and electron microscopic morphology . These are:
  - **Sarcomastigophora**, which includes flagellates and amoebae.
  - **Apicomplexa**, which includes intracellular parasites such as Plasmodium, Toxoplasma, and Cryptosporidium.
  - **Ciliophora**, which includes ciliates such as Paramecium, Stentor, and Vorticella.
  - **Microspora**, which includes spore-forming parasites such as Encephalitozoon and Nosema.
  - **Ascetosporea**, which includes parasites of marine invertebrates such as Myxozoa and Haplosporidia.
  - **Myxozoa**, which includes parasites of fish and aquatic worms such as Myxobolus and Ceratomyxa.
- Protozoa are important in many ecological and biomedical aspects, such as nutrient cycling, decomposition, symbiosis, pathogenesis, and evolution .

## Helminthes

- Helminthes are **worm-like parasites** that can infect humans and other animals .
- Helminthes are classified into three major groups based on their shape and

habitat :

- **Flukes (trematodes)**: flat, leaf-shaped worms that live in the liver, lungs, blood vessels, or intestines of the host.
- **Tapeworms (cestodes)**: segmented, ribbon-like worms that live in the intestines of the host and absorb nutrients from the digested food.
- **Roundworms (nematodes)**: cylindrical, thread-like worms that live in various tissues or organs of the host, such as the intestines, blood, lymphatic system, or subcutaneous tissues.
- Helminthes have complex life cycles that involve one or more intermediate hosts, such as snails, fish, insects, or mammals, where they undergo larval development before infecting the definitive host .
- Helminthes can cause various diseases in humans, such as schistosomiasis, cysticercosis, filariasis, ascariasis, hookworm infection, trichinosis, and onchocerciasis .
- Helminthes can be diagnosed by microscopic examination of stool, urine, blood, or tissue samples, or by serological tests that detect antibodies or antigens in the host's blood .
- Helminthes can be treated by anthelmintic drugs that kill or expel the worms from the host's body, such as praziquantel, albendazole, mebendazole, ivermectin, or diethylcarbamazine .
- Helminthes can be prevented by improving sanitation, hygiene, and access to clean water, as well as by avoiding contact with contaminated soil, water, food, or animals .

## Viral Structures

- Viruses are noncellular genetic elements that use a living cell for their replication and have an extracellular state.
- Viruses are ultramicroscopic particles containing nucleic acid (DNA or RNA) surrounded by a protein shell called a capsid .
- Some viruses have an outer envelope composed of lipids and proteins.
- Viruses are observed in a wide range of shapes and sizes, but these shapes are constant in a particular viral family.
- The basic structure of all viruses includes nucleic acid, capsid, and sometimes envelope.
- The nucleic acid of a virus contains the genetic information for the synthesis of proteins and replication.
- The capsid is a protein layer that forms a shell enclosing the genetic material of the virus.
- The envelope is a lipoprotein bilayer that encloses the capsid and may have spikes or glycoproteins for attachment to host cells.

- The two main classes of viruses are RNA viruses and DNA viruses, based on the type of nucleic acid they contain.
- Viruses can be classified into different types based on their shape, symmetry, presence or absence of envelope, and genome structure.
- Some of the common types of viruses are:
  - Helical viruses: These viruses have a capsid with a helical shape, like a coiled spring. The nucleic acid is wound around the inner surface of the capsid. Examples are tobacco mosaic virus, rabies virus, and Ebola virus.
  - Icosahedral viruses: These viruses have a capsid with a spherical shape, composed of 20 triangular faces. The nucleic acid is packed inside the capsid. Examples are poliovirus, adenovirus, and herpesvirus.
  - Enveloped viruses: These viruses have an envelope derived from the host cell membrane, which may contain viral proteins. The capsid is usually icosahedral or helical. Examples are influenza virus, HIV, and coronavirus.
  - Complex viruses: These viruses have a complex structure that does not fit into the above categories. They may have additional components, such as a tail, a head, or fibers. Examples are bacteriophages, poxviruses, and mimiviruses.
- The structure of viruses determines their infectivity, host range, antigenicity, and resistance to physical and chemical agents.

## Viral Multiplication

Viral multiplication is the process of producing new viruses using the host cell's machinery and resources. Viruses are obligate intracellular parasites, meaning they can only replicate inside living cells. Viruses have different types of genomes, such as DNA or RNA, single-stranded or double-stranded, linear or circular, etc. The type of genome determines the mode of replication and the enzymes involved.

The general steps of viral multiplication are:

1. **Attachment:** The virus binds to a specific receptor on the surface of the host cell. The receptor is usually a protein or a carbohydrate that is involved in normal cellular functions. The attachment is mediated by viral proteins called spikes or fibers that protrude from the viral capsid or envelope. The attachment determines the host range and tissue tropism of the virus, i.e., the types of cells and organisms that the virus can infect.
2. **Penetration:** The virus enters the host cell by one of the following mechanisms:
  - **Endocytosis:** The host cell membrane invaginates and forms a vesicle around the virus, which is then internalized into the cytoplasm.

This is common for enveloped viruses and some naked viruses.

- **Fusion:** The viral envelope fuses with the host cell membrane, releasing the viral capsid or nucleocapsid into the cytoplasm. This is common for some enveloped viruses, such as HIV and influenza.
  - **Injection:** The virus injects its genome into the host cell, leaving the capsid outside. This is common for some naked viruses, such as bacteriophages and plant viruses.
3. **Uncoating:** The viral genome is released from the capsid or nucleocapsid and becomes accessible for replication and transcription. This may involve the degradation of the capsid by viral or host enzymes, or the escape of the genome from the endocytic vesicle.
  4. **Biosynthesis:** The viral genome is replicated and transcribed to produce new copies of viral nucleic acids and proteins. Depending on the type of genome, this may occur in the cytoplasm or the nucleus of the host cell, and may involve different strategies and enzymes. For example, DNA viruses usually use the host cell's DNA polymerase to replicate their genome, while RNA viruses usually carry their own RNA polymerase to replicate their genome. The viral proteins are synthesized by the host cell's ribosomes, either from the viral genome directly or from viral mRNA transcripts.
  5. **Assembly:** The newly synthesized viral components are assembled into new viral particles. This may involve the formation of capsids around the viral genomes, the acquisition of envelopes from the host cell membrane, and the incorporation of viral spikes or fibers into the envelopes. The assembly may occur in the cytoplasm or the nucleus of the host cell, depending on the type of virus.
  6. **Release:** The new viral particles exit the host cell by one of the following mechanisms:
    - **Lysis:** The host cell membrane is ruptured by viral or host enzymes, releasing the viral particles into the extracellular environment. This is common for naked viruses and some enveloped viruses, such as herpesviruses. The lysis of the host cell usually results in its death.
    - **Budding:** The viral particles push out from the host cell membrane, acquiring envelopes as they do so. This is common for enveloped viruses, such as HIV and influenza. The budding of the viral particles may not kill the host cell immediately, but may cause gradual damage and dysfunction.
    - **Exocytosis:** The viral particles are packaged into vesicles and transported to the host cell membrane, where they are released by fusion of the vesicles with the membrane. This is common for some enveloped viruses, such as coronaviruses and paramyxoviruses. The exocytosis of the viral particles may not affect the host cell viability.

The viral multiplication cycle can be classified into two types based on the fate of the host cell and the viral genome:

- **Lytic cycle:** The virus replicates and produces new viral particles that

lyse the host cell and infect new cells. The viral genome does not integrate into the host genome and does not persist in the host cell. This is common for most viruses, such as bacteriophages and animal viruses.

- **Lysogenic cycle:** The virus integrates its genome into the host genome and becomes a prophage or a provirus. The viral genome remains dormant and does not produce new viral particles, but is replicated along with the host genome. The host cell is not lysed and continues

### **Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

- Microorganisms are used in industry to produce a variety of organic compounds including acids, growth stimulants, and enzymes.
- Microorganisms can be genetically modified or engineered to aid in large-scale production of desired products.
- Some examples of industrial chemicals and pharmaceuticals produced by microorganisms are:
  - Amino acids: Microbes make only the L-isomer of amino acids, which are essential for protein synthesis and human nutrition. Some amino acids, such as lysine and glutamic acid, are produced by fermentation of sugars or molasses using bacteria such as *Corynebacterium* or *Brevibacterium*.
  - Citric acid: Citric acid is widely used as a flavoring agent, preservative, and antioxidant in food and beverages. It is produced by the mold *Aspergillus niger*, which converts glucose or sucrose into citric acid in a submerged aerobic fermentation process.
  - Enzymes: Enzymes are biological catalysts that speed up chemical reactions. They have many applications in food, detergent, textile, paper, leather, and pharmaceutical industries. Some examples of enzymes produced by microorganisms are amylases, proteases, lipases, cellulases, and pectinases. These enzymes are usually secreted by fungi or bacteria, such as *Aspergillus*, *Bacillus*, or *Streptomyces*.
  - Antibiotics: Antibiotics are substances that inhibit the growth or kill other microorganisms. They are used to treat bacterial infections and prevent diseases. Most antibiotics are produced by fungi or bacteria, such as *Penicillium*, *Streptomyces*, or *Bacillus*. Some examples of antibiotics are penicillin, streptomycin, tetracycline, and erythromycin.
  - Hormones: Hormones are chemical messengers that regulate various physiological processes in animals and plants. They are used to treat hormonal disorders, enhance growth, and improve reproduction. Some examples of hormones produced by microorganisms are insulin, human growth hormone, and interferon. These hormones are usually produced by recombinant DNA technology, which involves inserting

the gene for the hormone into a suitable host cell, such as *Escherichia coli* or *Saccharomyces cerevisiae* .

- Biofuels: Biofuels are renewable sources of energy that can be derived from biomass, such as plants or microorganisms. They are used to reduce greenhouse gas emissions and dependence on fossil fuels. Some examples of biofuels produced by microorganisms are ethanol, biodiesel, biogas, and hydrogen. These biofuels are usually produced by fermentation, transesterification, or gasification of organic substrates, such as sugars, starches, oils, or wastes, using yeasts, bacteria, or algae .
- In the industrial process, microorganisms act like chemical factories. To be effective, they should liberate a large amount of a single product that can be efficiently isolated and purified. The microorganisms should be easy to maintain and cultivate, and should have genetic stability with infrequent mutations.
- The industrial process involves several steps, such as strain selection, media optimization, fermentation, downstream processing, and quality control. These steps aim to maximize the yield, quality, and safety of the product .

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells:

## # Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells

### ## Size, shape, and arrangement of bacterial cells

- Bacteria are microscopic organisms that can have various shapes and arrangements.
- The size of bacterial cells ranges from 0.2 to 10 micrometers ( $\mu\text{m}$ ) in diameter, and from 2 to 10 micrometers in length.
- The shape of bacterial cells can be classified into three main categories: cocci (spherical), bacilli (rod-shaped), and spirilla (spiral-shaped).
- The arrangement of bacterial cells depends on their pattern of division and attachment. Some common arrangements include:
  - Diplococci: pairs of cocci
  - Streptococci: chains of cocci
  - Staphylococci: grape-like clusters of cocci
  - Diplobacilli: pairs of bacilli
  - Streptobacilli: chains of bacilli
  - Palisades: parallel rows of bacilli
  - Spirilla: single or loose coils of spiral cells
  - Spirochetes: tight coils of spiral cells

### ## Structure and function of cells

- Cells are the basic units of life that perform various functions, such as metabolism, growth, and reproduction.
- Cells can be divided into two major types: prokaryotic and eukaryotic. Prokaryotic cells are simpler and lack a nucleus, while eukaryotic cells are more complex and have a nucleus.
- The structure and function of prokaryotic and eukaryotic cells can be compared as follows:

|                           |                                                            |                                                |  |
|---------------------------|------------------------------------------------------------|------------------------------------------------|--|
| Structure                 | Prokaryotic Cell                                           | Eukaryotic Cell                                |  |
| ---                       | ---                                                        | ---                                            |  |
| Nucleus                   | Absent                                                     | Present                                        |  |
| DNA                       | Circular, in cytoplasm                                     | Linear, in nucleus                             |  |
| Membrane-bound organelles | Absent                                                     | Present                                        |  |
| Cell wall                 | Present, made of peptidoglycan                             | Present in plants and fungi, made of cellulose |  |
| Plasma membrane           | Present, made of phospholipids and proteins                | Present, made of phospholipids and proteins    |  |
| Ribosomes                 | Present, smaller (70S)                                     | Present, larger (80S)                          |  |
| Cytoskeleton              | Absent                                                     | Present                                        |  |
| Flagella                  | Present, made of flagellin                                 | Present in some cells, made of microtubules    |  |
| Pili                      | Present, for attachment and conjugation                    | Absent                                         |  |
| Capsule                   | Present in some cells, for protection and adhesion         | Absent                                         |  |
| Endospore                 | Present in some cells, for survival under harsh conditions | Absent                                         |  |
| Plasmid                   | Present in some cells, for extra genetic information       | Absent                                         |  |

### Functional Anatomy of Prokaryotic and Eukaryotic Cells

- Prokaryotic and eukaryotic cells are the two basic types of cells that make up all living organisms .
- Prokaryotic cells are simpler and smaller than eukaryotic cells, and lack a nucleus and other membrane-bound organelles .
- Eukaryotic cells are more complex and larger than prokaryotic cells, and have a nucleus and other membrane-bound organelles that perform specific functions .
- The main differences between prokaryotic and eukaryotic cells are summarized in the table below:

| Feature       | Prokaryotic cells                                                   | Eukaryotic cells                                                                        |
|---------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Size          | 0.1-5 micrometers                                                   | 10-100 micrometers                                                                      |
| Nucleus       | Absent                                                              | Present                                                                                 |
| DNA           | Circular, in nucleoid region                                        | Linear, in nucleus                                                                      |
| Histones      | Absent                                                              | Present                                                                                 |
| Organelles    | Absent (except ribosomes)                                           | Present (e.g. mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, etc.) |
| Cell wall     | Present (peptidoglycan in bacteria, pseudopeptidoglycan in archaea) | Absent (except in plants, fungi, and some protists)                                     |
| Cell division | Binary fission                                                      | Mitosis or meiosis                                                                      |
| Ribosomes     | 70S                                                                 | 80S                                                                                     |

- The size, shape, and arrangement of bacterial cells vary widely, and can be used to identify and classify different types of bacteria .
- Some common shapes of bacterial cells are:
  - Coccus (spherical)
  - Bacillus (rod-shaped)
  - Coccobacillus (short and plump)
  - Vibrio (curved rod)
  - Spirillum (rigid spiral)
  - Spirochete (flexible spiral)
  - Pleomorphic (variable shape)
- Some common arrangements of bacterial cells are:
  - Single (individual cells)
  - Diplo (pairs of cells)
  - Strepto (chains of cells)
  - Staphylo (clusters of cells)
  - Tetrad (groups of four cells)
  - Sarcina (cubical packets of eight or more cells)
  - Palisade (parallel rows of cells)
- The structure and function of prokaryotic and eukaryotic cells are determined by the interactions of their components, such as membranes, cytoplasm, ribosomes, DNA, organelles, and cell walls .
- The following diagram shows the basic structure of a prokaryotic cell and a eukaryotic cell:

prokaryotic and eukaryotic cells

## Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers ( $\mu\text{m}$ ), which are one-thousandth of a millimeter. The average size of a bacterial cell varies between 0.2 and 2.0  $\mu\text{m}$  .
- Bacterial cells have different shapes and arrangements that help them adapt to different environments and functions. The most common shapes are:
  - Cocci: spherical or oval cells that can be arranged in pairs (diplococci), chains (streptococci), clusters (staphylococci), tetrads (groups of four), sarcinae (cubical packets of eight), or other patterns .
  - Bacilli: rod-shaped or cylindrical cells that can be arranged in singles, pairs (diplobacilli), chains (streptobacilli), or palisades (side-by-side or V-shaped patterns) .



- Spirilla: rigid, helical or spiral-shaped cells that can be arranged in singles or pairs .
- Spirochetes: flexible, corkscrew-shaped cells that can move by rotating their axial filaments .
- Vibrios: comma-shaped or curved cells that can be arranged in singles or pairs .
- Coccobacilli: oval or elongated cells that are intermediate between cocci and bacilli .
- Filamentous: long, thin, or branched cells that can form networks or mats .
- Some bacterial cells have special shapes or structures that are not common, such as:
  - Star-shaped: cells that have projections or rays from their central body, such as *Stella*.
  - Square-shaped: cells that have flat sides and sharp corners, such as *Haloarcula*.
  - Triangular-shaped: cells that have three sides and angles, such as *Haloquadratum*.
  - Nanobacteria: cells that are extremely small, ranging from 50 to 500 nm, and are controversial as to whether they are living or not.

## Size, shape, and arrangement of bacterial cells

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  - Bacilli: rod-shaped or cylindrical cells that can be arranged in singles, pairs (diplobacilli), chains (streptobacilli), or palisades (side-by-side or parallel rows) .
  - Spirilla: rigid, helical or spiral-shaped cells that can be arranged in singles or pairs .
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  - Coccobacilli: oval or elongated cells that are intermediate between cocci and bacilli .
  - Filamentous: long, thin, or branched cells that can form chains or

networks .

- The shape and arrangement of bacterial cells are determined by their cell wall structure, cytoskeleton, and environmental factors. Some bacteria can change their shape and arrangement in response to different conditions, such as nutrient availability, temperature, pH, or stress .
- The shape and arrangement of bacterial cells can affect their motility, attachment, colonization, pathogenicity, and resistance to antibiotics and immune system .

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### Arrangement of Bacterial Cells

Bacteria are microscopic, single-celled organisms that come in different shapes and sizes. They can be classified based on their shape, cell wall structure, and arrangement. The arrangement of bacterial cells depends on how they divide and how they stick together after division.

- **Shape:** Bacteria can have one of three basic shapes: coccus (spherical), bacillus (rod-shaped), or spiral (curved or twisted). Some bacteria can change their shape depending on the environment or the stage of their life cycle. For example, some bacteria can form spores, which are dormant and resistant forms of bacteria that can survive harsh conditions.
- **Cell wall structure:** Bacteria can have different types of cell walls, which affect their staining properties and their susceptibility to antibiotics. The most common classification is based on the Gram stain, which divides bacteria into Gram-positive and Gram-negative. Gram-positive bacteria have a thick layer of peptidoglycan, a polymer of sugars and proteins, in their cell wall, which retains the purple dye of the Gram stain. Gram-negative bacteria have a thin layer of peptidoglycan and an outer membrane of lipopolysaccharides, which does not retain the dye and appears pink after counterstaining.
- **Arrangement:** Bacteria can have different patterns of arrangement based on how they divide and how they stick together after division. The most common arrangements are diplo, staphylo, strepto, tetrad, and sarcina. These arrangements can be applied to the different bacterial shapes. For example, diplococci are pairs of spherical cells, streptobacilli are chains of rod-shaped cells, and sarcinae are cubic packets of eight cells .

- Diplo: This arrangement consists of two cells that remain attached after division. Diplococci are pairs of cocci, and diplobacilli are pairs of bacilli. Examples of diplococci are *Neisseria gonorrhoeae*, which causes gonorrhea, and *Streptococcus pneumoniae*, which causes pneumonia .
- Staphylo: This arrangement consists of clusters of cells that divide in multiple planes and form grape-like structures. Staphylococci are clusters of cocci, and staphylobacilli are clusters of bacilli. Examples of staphylococci are *Staphylococcus aureus*, which causes skin infections and food poisoning, and *Staphylococcus epidermidis*, which is part of the normal skin flora .
- Strepto: This arrangement consists of chains of cells that divide in one plane and remain attached end to end. Streptococci are chains of cocci, and streptobacilli are chains of bacilli. Examples of streptococci are *Streptococcus pyogenes*, which causes strep throat and rheumatic fever, and *Streptococcus mutans*, which causes dental caries .
- Tetrad: This arrangement consists of four cells that divide in two planes and form a square. Tetrads are squares of cocci, and tetrads bacilli are squares of bacilli. Examples of tetrads are *Micrococcus luteus*, which is part of the normal skin flora, and *Micrococcus roseus*, which produces a pink pigment .
- Sarcina: This arrangement consists of eight cells that divide in three planes and form a cube. Sarcinae are cubes of cocci, and sarcinabacilli are cubes of bacilli. Examples of sarcinae are *Sarcina ventriculi*, which is found in the stomach of animals, and *Sarcina maxima*, which is one of the largest bacteria known .

: **Prokaryotic cells (article) | Khan Academy** What Is Arrangement in Microbiology? | Sciencing

Different Size, Shape and Arrangement of Bacterial Cells

Bacterial Cells: Size, Shape and Arrangement (With Diagram)

2.1: Sizes, Shapes, and Arrangements of Bacteria

### Structure and function of cells

- Cells are the basic functional units that make up all living organisms. They have different shapes, sizes, and arrangements depending on their functions and types .
- Cells are enclosed by a cell membrane that separates the inside and the outside of the cell, and controls what goes in and comes out .
- Cells contain different functional structures which are collectively called organelles, and they are involved in various cellular functions .
- Cells can be divided into two major types: prokaryotic and eukaryotic. Prokaryotic cells are simpler and smaller than eukaryotic cells, and do not

have a nucleus or other membrane-bound organelles. Eukaryotic cells are more complex and larger than prokaryotic cells, and have a nucleus and other membrane-bound organelles .

- Some of the common organelles found in eukaryotic cells are:
  - Nucleus: It contains the genetic material (DNA) of the cell and controls its activities .
  - Mitochondria: They are the sites of cellular respiration, where energy is produced from glucose and oxygen .
  - Ribosomes: They are the sites of protein synthesis, where amino acids are joined together to form proteins .
  - Endoplasmic reticulum: It is a network of membranes that transports materials within the cell and helps in the synthesis of lipids and proteins .
  - Golgi apparatus: It is a stack of flattened membranes that modifies, sorts, and packages proteins and other molecules for export or storage .
  - Lysosomes: They are vesicles that contain digestive enzymes that break down waste materials and foreign substances in the cell .
  - Cytoskeleton: It is a network of protein filaments that provides shape, support, and movement to the cell .
- Some of the common organelles found in prokaryotic cells are:
  - Cell wall: It is a rigid layer that surrounds the cell membrane and provides protection and support to the cell .
  - Cytoplasm: It is the fluid-filled space inside the cell that contains the genetic material, ribosomes, and other molecules .
  - Plasmids: They are small circular pieces of DNA that can replicate independently and carry genes for antibiotic resistance or other traits .
  - Flagella: They are long whip-like structures that help the cell to move .
  - Pili: They are short hair-like structures that help the cell to attach to other cells or surfaces .

Hello, I am Sydney, your AI assistant. I can help you with your study material on Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism.

Here is a summary of the main points of this unit:

### **Catabolic and anabolic reactions**

- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. An example of a catabolic reaction is glycolysis, which breaks down glucose into pyruvate and produces ATP and NADH.

- Anabolic reactions are those that build up complex molecules from simpler ones, consuming energy in the process. An example of an anabolic reaction is gluconeogenesis, which synthesizes glucose from non-carbohydrate sources such as lactate and amino acids, using ATP and NADH.
- Enzymes are biological catalysts that speed up the rate of chemical reactions by lowering the activation energy. Enzymes are highly specific for their substrates and can be regulated by various factors such as temperature, pH, inhibitors, activators, and feedback mechanisms.
- Energy production is the process of converting the chemical energy stored in the bonds of organic molecules into a more usable form, such as ATP, which is the universal energy currency of the cell. Energy production can occur through different pathways, depending on the availability of oxygen and the type of substrate.

## Carbohydrate metabolism

- Carbohydrate metabolism is the set of reactions that involve the breakdown and synthesis of carbohydrates, such as glucose, glycogen, starch, and cellulose. Carbohydrate metabolism can be divided into three main stages: glycolysis, the citric acid cycle, and oxidative phosphorylation.
- Glycolysis is the first stage of carbohydrate metabolism, which occurs in the cytoplasm of the cell. It converts one molecule of glucose into two molecules of pyruvate, generating two molecules of ATP and two molecules of NADH. Glycolysis can occur in the presence or absence of oxygen, and can be followed by different pathways depending on the cellular conditions.
- The citric acid cycle, also known as the Krebs cycle or the tricarboxylic acid cycle, is the second stage of carbohydrate metabolism, which occurs in the mitochondrial matrix. It oxidizes the acetyl-CoA derived from pyruvate or fatty acids, producing two molecules of CO<sub>2</sub>, one molecule of ATP, three molecules of NADH, and one molecule of FADH<sub>2</sub> per cycle. The citric acid cycle requires oxygen and is the main source of carbon dioxide and reduced coenzymes for oxidative phosphorylation.
- Oxidative phosphorylation is the third and final stage of carbohydrate metabolism, which occurs in the inner mitochondrial membrane. It uses the electrons from NADH and FADH<sub>2</sub> to drive the synthesis of ATP from ADP and Pi, through a series of protein complexes called the electron transport chain and a proton gradient across the membrane. Oxidative phosphorylation is the most efficient way of producing ATP, generating about 28 molecules of ATP per molecule of glucose.

## Lipid and protein catabolism

- Lipid catabolism is the process of breaking down lipids, such as triglycerides, into glycerol and fatty acids, which can then be used as energy sources or substrates for other metabolic pathways. Lipid catabolism can occur through two main mechanisms: lipolysis and beta-oxidation.

- Lipolysis is the hydrolysis of triglycerides into glycerol and fatty acids, which occurs in the cytoplasm of adipose tissue and other cells. Lipolysis is stimulated by hormones such as glucagon, epinephrine, and cortisol, and inhibited by insulin. Glycerol can be converted into glucose by gluconeogenesis, while fatty acids can be transported to the mitochondria for beta-oxidation.
- Beta-oxidation is the oxidation of fatty acids into acetyl-CoA, which occurs in the mitochondrial matrix. Beta-oxidation involves four steps: dehydrogenation, hydration, oxidation, and thiolysis, which are repeated until the fatty acid chain is completely cleaved. Beta-oxidation produces one molecule of FADH<sub>2</sub> and one molecule of NADH per cycle, as well as acetyl-CoA, which can enter the citric acid cycle or be used for ketone body synthesis.
- Protein catabolism is the process of breaking down proteins into amino acids, which can then be used as energy sources or substrates for other metabolic pathways. Protein catabolism can occur through two main mechanisms: proteol

## Catabolic and Anabolic Reactions

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. An example of a catabolic reaction is glycolysis, which breaks down glucose into pyruvate and produces ATP .
- Anabolic reactions are those that build up complex molecules from simpler ones, using energy in the process. An example of an anabolic reaction is protein synthesis, which links amino acids together to form proteins .
- Catabolic and anabolic reactions are often coupled, meaning that the energy released by catabolic reactions is used to power anabolic reactions. A common molecule that mediates this coupling is ATP, which is produced by catabolic reactions and consumed by anabolic reactions .
- Catabolic and anabolic reactions are essential for the growth, maintenance, and repair of cells and tissues. They also regulate the balance of energy and matter in the body .

## Enzymes

Enzymes are biological catalysts that speed up the rate of biochemical reactions by lowering the activation energy. Enzymes are highly specific for their substrates and can be regulated by various factors such as temperature, pH, inhibitors, activators, and feedback mechanisms. Enzymes can be classified into two main types based on the direction of the reaction they catalyze: anabolic and catabolic enzymes.

## Anabolic Enzymes

Anabolic enzymes are involved in the synthesis of larger complex molecules from smaller units. Anabolic reactions are also called biosynthetic reactions because they produce new biomolecules that are essential for the growth and maintenance of cells. Anabolic reactions consume energy, usually in the form of ATP, and are endergonic (positive  $\Delta G$ ). An example of an anabolic reaction is photosynthesis, where plants make glucose molecules from carbon dioxide and water using light energy. Some examples of anabolic enzymes are:

- DNA polymerase: synthesizes DNA from nucleotides
- RNA polymerase: synthesizes RNA from nucleotides
- Ribosome: synthesizes proteins from amino acids
- Hexokinase: adds a phosphate group to glucose
- RuBisCO: fixes carbon dioxide into organic molecules

## Catabolic Enzymes

Catabolic enzymes are involved in the breakdown of larger complex molecules into smaller units. Catabolic reactions are also called degradative reactions because they degrade biomolecules that are no longer needed or that can be used as energy sources. Catabolic reactions release energy, usually in the form of ATP, and are exergonic (negative  $\Delta G$ ). An example of a catabolic reaction is cellular respiration, where glucose molecules are oxidized to carbon dioxide and water, releasing energy. Some examples of catabolic enzymes are:

- Amylase: breaks down starch into glucose
- Lipase: breaks down fats into glycerol and fatty acids
- Protease: breaks down proteins into amino acids
- Pyruvate kinase: removes a phosphate group from pyruvate
- Cytochrome c oxidase: transfers electrons from cytochrome c to oxygen

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**Energy production for the notes of the Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism in the subject of REMEDIAL BIOLOGY-II KCS**

- Energy production is the process of converting different sources of energy, such as chemical, light, or mechanical, into a usable form, such as ATP, the universal energy currency of cells.
- Energy production involves two types of metabolic pathways: catabolic and anabolic.
- Catabolic pathways are those that break down large molecules into smaller ones, releasing energy in the process. Examples of catabolic pathways are glycolysis, the citric acid cycle, and oxidative phosphorylation, which are involved in carbohydrate metabolism; beta-oxidation, which is involved in lipid metabolism; and proteolysis, which is involved in protein metabolism.
- Anabolic pathways are those that synthesize large molecules from smaller ones, requiring energy in the process. Examples of anabolic pathways are gluconeogenesis, the synthesis of glucose from non-carbohydrate sources; lipogenesis, the synthesis of fatty acids and triglycerides; and protein synthesis, the synthesis of amino acids and polypeptides.
- Enzymes are biological catalysts that speed up the rate of chemical reactions in metabolic pathways. Enzymes lower the activation energy of the reactions, making them more likely to occur. Enzymes are highly specific for their substrates and products, and are regulated by various factors, such as temperature, pH, inhibitors, and activators.
- Energy production mechanism is the process of coupling the energy released from catabolic pathways to the energy required for anabolic pathways. The most common energy production mechanism in cells is the chemiosmotic coupling, which involves the generation of a proton gradient across a membrane by the electron transport chain, and the use of this gradient to drive the synthesis of ATP by the enzyme ATP synthase.
- Metabolic diversity is the variation in the types and pathways of energy production among different organisms, depending on their environmental conditions and evolutionary history. Some examples of metabolic diversity are:
  - Aerobic respiration, which uses oxygen as the final electron acceptor and produces the most ATP per glucose molecule.
  - Anaerobic respiration, which uses other molecules, such as nitrate, sulfate, or carbon dioxide, as the final electron acceptor and produces less ATP per glucose molecule.
  - Fermentation, which does not use an electron transport chain and produces the least ATP per glucose molecule, but allows the regeneration of NAD<sup>+</sup> for glycolysis.
  - Photosynthesis, which uses light energy to convert carbon dioxide and water into glucose and oxygen, and involves two stages: the light



reactions and the Calvin cycle.

- Chemolithotrophy, which uses inorganic molecules, such as hydrogen, sulfur, or iron, as the source of electrons and energy, and carbon dioxide as the source of carbon.
- Chemoorganotrophy, which uses organic molecules, such as glucose, as the source of electrons, energy, and carbon.
- Pathways of energy use are the ways that cells use the energy produced from different sources to perform various functions, such as growth, maintenance, movement, signaling, and reproduction. Some examples of pathways of energy use are:
  - Biosynthesis, which uses energy to synthesize the building blocks of macromolecules, such as nucleotides, amino acids, and fatty acids.
  - Transport, which uses energy to move molecules and ions across membranes, such as the sodium-potassium pump, the proton pump, and the glucose transporter.
  - Mechanical work, which uses energy to generate force and movement, such as the actin-myosin interaction in muscle contraction, the flagellar rotation in bacterial motility, and the kinesin-dynein interaction in intracellular transport.
  - Electrical work, which uses energy to generate and maintain electrical potentials across membranes, such as the action potential in nerve cells, the resting potential in muscle cells, and the membrane potential in plant cells.
  - Heat production, which uses energy to maintain a constant body temperature, such as the brown fat tissue in mammals, the shivering in birds, and the thermogenesis in plants.
- Integration of metabolism is the coordination and regulation of the various metabolic pathways and energy production mechanisms in response to the changing needs and conditions of the cell and the organism. Integration of metabolism involves various levels of control, such as:
  - Allosteric regulation, which involves the binding of a molecule to an enzyme,

## Carbohydrate metabolism

Carbohydrate metabolism is the set of biochemical processes that involve the formation, breakdown, and interconversion of carbohydrates in living organisms. Carbohydrates are organic molecules that consist of carbon, hydrogen, and oxygen atoms. They can be classified into simple sugars (monosaccharides) and complex sugars (polysaccharides).

Some of the main functions of carbohydrates are:

- To provide energy for cellular activities through the oxidation of glucose
- To store energy in the form of glycogen in animals and starch in plants
- To serve as structural components of cell walls (cellulose) and extracellular matrices (glycosaminoglycans)

- To participate in cell signaling and recognition through glycoproteins and glycolipids

The liver is the main organ that regulates carbohydrate metabolism, as it can store and distribute glucose to other tissues, especially the brain and red blood cells, which depend on glucose as their primary fuel. The blood glucose level is maintained within a narrow range by the action of three hormones: insulin, glucagon, and epinephrine.

Insulin is secreted by the pancreas when the blood glucose level is high, and it stimulates the uptake of glucose by the cells, the conversion of glucose to glycogen (glycogenesis), and the synthesis of fatty acids (lipogenesis). Glucagon is secreted by the pancreas when the blood glucose level is low, and it stimulates the breakdown of glycogen (glycogenolysis), the production of glucose from non-carbohydrate sources (gluconeogenesis), and the release of fatty acids (lipolysis). Epinephrine is secreted by the adrenal glands in response to stress, and it enhances the effects of glucagon and inhibits the effects of insulin.

The main pathways of carbohydrate metabolism are:

- Glycolysis: the breakdown of glucose to pyruvate, which can be further oxidized to acetyl-CoA in the presence of oxygen (aerobic glycolysis) or converted to lactate in the absence of oxygen (anaerobic glycolysis). Glycolysis occurs in the cytoplasm of all cells and produces a net gain of 2 ATP and 2 NADH molecules per glucose molecule.
- Citric acid cycle (Krebs cycle): the oxidation of acetyl-CoA to carbon dioxide and water, which occurs in the mitochondrial matrix of eukaryotic cells and the cytoplasm of prokaryotic cells. The citric acid cycle produces a net gain of 2 ATP, 6 NADH, and 2 FADH<sub>2</sub> molecules per acetyl-CoA molecule.
- Oxidative phosphorylation: the transfer of electrons from NADH and FADH<sub>2</sub> to oxygen, which occurs in the inner mitochondrial membrane of eukaryotic cells and the plasma membrane of prokaryotic cells. Oxidative phosphorylation generates a proton gradient across the membrane, which drives the synthesis of ATP by the enzyme ATP synthase. Oxidative phosphorylation produces a net gain of about 28 ATP molecules per glucose molecule.
- Pentose phosphate pathway: the conversion of glucose-6-phosphate to ribose-5-phosphate and NADPH, which occurs in the cytoplasm of all cells. The pentose phosphate pathway provides the precursors for nucleotide synthesis and the reducing power for biosynthetic reactions.
- Glycogenesis: the synthesis of glycogen from glucose, which occurs in the cytoplasm of liver and muscle cells. Glycogenesis is stimulated by insulin and inhibited by glucagon and epinephrine.
- Glycogenolysis: the breakdown of glycogen to glucose, which occurs in the cytoplasm of liver and muscle cells. Glycogenolysis is stimulated by glucagon and epinephrine and inhibited by insulin.
- Gluconeogenesis: the production of glucose from non-carbohydrate

sources, such as lactate, pyruvate, glycerol, and amino acids, which occurs mainly in the cytoplasm and mitochondria of liver cells. Gluconeogenesis is stimulated by glucagon and epinephrine and inhibited by insulin.

Carbohydrate metabolism is influenced by various factors, such as dietary intake, hormonal regulation, enzyme activity, genetic variation, and environmental conditions. Carbohydrate metabolism is also integrated with other metabolic pathways, such as lipid and protein metabolism, to maintain the homeostasis of energy and biomolecules in the body.

### **Lipid and Protein Catabolism**

Lipids and proteins are important macromolecules that serve as energy sources and structural components for living cells. They can be broken down into smaller molecules through catabolic reactions that release energy and feed into the central metabolic pathways.

- Lipid catabolism involves the hydrolysis of triglycerides (fats) into glycerol and fatty acids by lipases, enzymes that are secreted by some microorganisms or released from adipose tissue in animals .
- Glycerol can be converted into dihydroxyacetone phosphate (DHAP), an intermediate of glycolysis, by a series of enzymatic reactions .
- Fatty acids can be activated by coenzyme A (CoA) and transported into the mitochondrial matrix by carnitine shuttle, where they undergo beta-oxidation, a cyclic process that cleaves two-carbon units from the acyl-CoA chain and produces acetyl-CoA, NADH, and FADH<sub>2</sub> .
- Acetyl-CoA can enter the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle) and generate more NADH, FADH<sub>2</sub>, and GTP, which can be used for oxidative phosphorylation and ATP synthesis .
- Protein catabolism involves the proteolysis of proteins into amino acids by proteases, enzymes that are secreted by some microorganisms or released from lysosomes in animal cells .
- Amino acids can be transported into the cytoplasm and deaminated, a process that removes the amino group and produces ammonia and keto acids .
- Ammonia can be excreted as waste or converted into urea or uric acid in some animals, or assimilated into other nitrogen-containing compounds in some microorganisms .
- Keto acids can be converted into pyruvate, acetyl-CoA, or other intermediates of the citric acid cycle, depending on their chemical structures .
- Pyruvate and acetyl-CoA can enter the citric acid cycle and generate more NADH, FADH<sub>2</sub>, and GTP, which can be used for oxidative phosphorylation and ATP synthesis .

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Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information that I found from the web:

## Energy production mechanism

- Energy production is the process of converting energy from one form to another in living systems.
- Energy comes to living systems through electrons occupying high energy states, either from food (respiratory chains) or from light (photosynthesis).
- Energy is exchanged between living systems and their surroundings as they use energy from the sun to perform photosynthesis or consume energy-storing molecules and release energy to the environment by doing work and releasing heat.
- Energy is subject to physical laws, such as the first law of thermodynamics, which states that energy cannot be created or destroyed, only transformed, and the second law of thermodynamics, which states that the entropy of the universe increases in every spontaneous process.
- The primary energy-releasing pathways of metabolism involve the breakdown and synthesis of carbohydrates, lipids, and amino acids.
- Carbohydrates are the most important energy source for most living organisms. They are broken down into glucose, which is then oxidized through glycolysis, the citric acid cycle, and the electron transport chain to produce ATP, the universal energy currency of the cell.
- Lipids are also a major source of energy, especially for animals. They are stored as triglycerides in adipose tissue and can be mobilized when glucose levels are low. They are broken down into glycerol and fatty acids, which can enter glycolysis and the citric acid cycle, respectively, to generate ATP.
- Amino acids are the building blocks of proteins, but they can also be used as an energy source when carbohydrates and lipids are insufficient. They are deaminated to remove the amino group, which is excreted as urea or ammonia, and the remaining carbon skeleton can enter glycolysis or the citric acid cycle to produce ATP.
- Energy stored in organic molecules can be passed to other organisms in the ecosystem when those organisms eat plants (or eat other organisms that have previously eaten plants). This is called the flow of energy in the food chain or food web.
- Different organisms have different metabolic pathways to use energy for various purposes, such as growth, maintenance, reproduction, movement,

and defense. Some examples of metabolic diversity are anaerobic respiration, fermentation, chemosynthesis, and photosynthesis.

- The integration of metabolism is the coordination and regulation of the various metabolic pathways to maintain homeostasis and respond to environmental changes. Some examples of integration of metabolism are hormonal control, feedback inhibition, allosteric regulation, and substrate-level phosphorylation.

## Metabolic Diversity and Pathways of Energy Use

- Metabolic diversity mainly refers to the different metabolic strategies that organisms have evolved to obtain energy.
- Metabolic pathways are series of connected chemical reactions that feed one another and convert starting molecules into products.
- Metabolic pathways can be broadly divided into two categories based on their effects:
  - Catabolic pathways break down complex molecules into simpler ones and release energy. Examples are glycolysis, Krebs cycle, and oxidative phosphorylation.
  - Anabolic pathways synthesize complex molecules from simpler ones and require energy. Examples are photosynthesis, gluconeogenesis, and protein synthesis.
- Metabolic pathways evolved among prokaryotes before eukaryotes arose as the result of their interaction and coevolution with changing physico-chemical environmental conditions.
- Metabolic diversity is reflected in the different types of energy sources and electron acceptors that organisms can use:
  - Energy sources can be organic (such as glucose) or inorganic (such as hydrogen or ammonia). Organisms that use organic compounds as energy sources are called organotrophs, while those that use inorganic compounds are called lithotrophs.
  - Electron acceptors can be oxygen (aerobic respiration), nitrate (denitrification), sulfate (sulfate reduction), carbon dioxide (methanogenesis), or other compounds. Organisms that use oxygen as the electron acceptor are called aerobes, while those that use other compounds are called anaerobes.
- Metabolic diversity is also reflected in the different metabolic pathways that organisms can use to produce ATP, the universal energy currency of cells :
  - The phosphagen system (ATP-PC system) for immediate energy. This system uses creatine phosphate (CP) to regenerate ATP from ADP in a single enzymatic reaction. This system is used for short bursts of high-intensity activity, such as sprinting or lifting weights.
  - The glycolytic system (anaerobic glycolysis) for short-term energy. This system breaks down glucose into pyruvate and produces two

- ATP molecules per glucose molecule. This system is used for moderate-intensity activity, such as jogging or cycling.
- The oxidative system (aerobic respiration) for long-term energy. This system involves the complete oxidation of glucose, fatty acids, or amino acids into carbon dioxide and water, and produces 36 to 38 ATP molecules per glucose molecule. This system is used for low-intensity activity, such as walking or resting.
- Metabolic diversity allows organisms to adapt to different environmental conditions and to exploit different ecological niches .
- Metabolic diversity also influences the biogeochemical cycles of elements, such as carbon, nitrogen, sulfur, and iron, by transforming their chemical forms and availability .
- Metabolic diversity is a result of evolutionary processes that have shaped the metabolic capabilities of organisms over billions of years .

### **Integration of metabolism**

- Metabolism is the sum of all the chemical reactions that take place in a living cell to maintain its structure and function.
- Metabolism can be divided into two major types: catabolism and anabolism.
- Catabolism is the breakdown of complex molecules into simpler ones, releasing energy in the process. Examples of catabolic reactions are glycolysis, lipid oxidation, and protein degradation.
- Anabolism is the synthesis of complex molecules from simpler ones, requiring energy in the process. Examples of anabolic reactions are gluconeogenesis, lipid synthesis, and protein synthesis.
- Metabolism is not a random process, but a highly coordinated and regulated one. Different metabolic pathways are interconnected and influenced by various factors, such as the availability of substrates, the demand for energy, the hormonal status, and the environmental conditions.
- Integration of metabolism refers to the harmonious functioning of different metabolic pathways in different tissues and organs to maintain the homeostasis of the whole organism.
- Integration of metabolism is achieved by several mechanisms, such as allosteric regulation, covalent modification, transcriptional regulation, and hormonal regulation.
- Allosteric regulation is the binding of a molecule to an enzyme at a site other than the active site, affecting its activity. Allosteric regulators can be either activators or inhibitors. For example, ATP is an allosteric inhibitor of phosphofructokinase, a key enzyme in glycolysis, while AMP is an allosteric activator of the same enzyme.
- Covalent modification is the addition or removal of a chemical group to or from an enzyme, affecting its activity. Covalent modifications are usually reversible and mediated by specific enzymes. For example, phosphorylation is a common covalent modification that can activate or inhibit

enzymes, depending on the site and the enzyme. Phosphorylation is catalyzed by kinases and reversed by phosphatases.

- Transcriptional regulation is the control of gene expression by altering the rate of transcription of DNA into RNA. Transcriptional regulation is influenced by various factors, such as the binding of transcription factors, the methylation of DNA, and the acetylation of histones. For example, the expression of genes involved in lipid synthesis is increased by the transcription factor SREBP, which is activated by low cholesterol levels in the cell.
- Hormonal regulation is the control of metabolism by hormones, which are chemical messengers secreted by endocrine glands and transported by the blood to target tissues. Hormones can affect metabolism by binding to specific receptors on the cell membrane or inside the cell, triggering a cascade of events that alter the activity or expression of enzymes. For example, insulin is a hormone that stimulates glucose uptake and anabolic pathways, while glucagon is a hormone that stimulates glucose release and catabolic pathways.

## Unit 4 - Energy Utilization

### Structure of mitochondria

- Mitochondria are small, rod-shaped organelles found in most eukaryotic cells, such as plants and animals .
- They have two membranes, an outer membrane and an inner membrane, made of phospholipids .
- The outer membrane is smooth and surrounds the organelle, while the inner membrane is folded into projections called cristae .
- The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the matrix .
- The matrix contains mitochondrial DNA, ribosomes, enzymes, and other molecules involved in energy production .
- The cristae increase the surface area of the inner membrane, where the electron transport chain and oxidative phosphorylation occur .
- Mitochondria are the sites of cellular respiration, the process of converting organic molecules into ATP, the main energy currency of the cell .
- Mitochondria can vary in number, shape, and size depending on the cell type and energy demand .
- Mitochondria have their own genetic material and can replicate independently of the cell cycle.
- Mitochondria are thought to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells in a process called endosymbiosis.

## Cellular respiration

- Cellular respiration is the process of breaking down organic molecules, such as glucose, into carbon dioxide and water, and releasing energy in the form of ATP .
- Cellular respiration can be divided into four stages: glycolysis, formation of acetyl co-A, Krebs cycle, and electron transport chain and oxidative phosphorylation .
- Glycolysis is the first stage of cellular respiration, and it occurs in the cytoplasm of the cell. It involves splitting a molecule of glucose into two molecules of pyruvate, and producing two molecules of ATP and two molecules of NADH (a high-energy electron carrier) .
- Formation of acetyl co-A is the second stage of cellular respiration, and it occurs in the matrix of the mitochondria. It involves converting each molecule of pyruvate into a molecule of acetyl co-A, and releasing one molecule of carbon dioxide and one molecule of NADH .
- Krebs cycle, also known as the citric acid cycle or the tricarboxylic acid cycle, is the third stage of cellular respiration, and it occurs in the matrix of the mitochondria. It involves a series of reactions that use acetyl co-A to produce two molecules of carbon dioxide, one molecule of ATP, three molecules of NADH, and one molecule of FADH<sub>2</sub> (another high-energy electron carrier) per cycle .
- Electron transport chain and oxidative phosphorylation are the fourth and final stage of cellular respiration, and they occur in the inner membrane of the mitochondria. They involve transferring the electrons from NADH and FADH<sub>2</sub> to a series of protein complexes and electron carriers, and using the energy released to pump protons (H<sup>+</sup>) from the matrix to the intermembrane space, creating a proton gradient . The protons then flow back to the matrix through a protein called ATP synthase, which uses the energy to synthesize ATP from ADP and phosphate. This process is called oxidative phosphorylation because oxygen is the final electron acceptor and water is the final product .
- The total amount of ATP produced by cellular respiration from one molecule of glucose is about 36 to 38 molecules, depending on the efficiency of the process

## Energy Utilization

- Energy utilization is the process of converting chemical energy stored in organic molecules into other forms of energy that can be used by cells for various functions.
- Energy utilization involves a series of biochemical reactions that take place in different cellular compartments, such as the cytoplasm and the mitochondria.
- Energy utilization is mainly based on the oxidation of carbohydrates, such



as glucose, which can be derived from the digestion of food or the breakdown of glycogen.

- The oxidation of glucose involves three major stages: glycolysis, the citric acid cycle (or Krebs cycle), and the electron transport chain (or oxidative phosphorylation) .
- Each stage of glucose oxidation produces different products and releases different amounts of energy, some of which are captured in the form of adenosine triphosphate (ATP), the primary energy currency of all cells.

## Structure of mitochondria

- Mitochondria are organelles that are found in most eukaryotic cells, except for red blood cells.
- Mitochondria are the sites of the citric acid cycle and the electron transport chain, which are the main sources of ATP production in aerobic organisms.
- Mitochondria have a double-membrane structure, consisting of an outer membrane and an inner membrane.
- The outer membrane is smooth and permeable to small molecules, while the inner membrane is folded into numerous projections called cristae, which increase the surface area for the electron transport chain.
- The space between the two membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix.
- The matrix contains various enzymes, coenzymes, and molecules involved in the citric acid cycle, as well as mitochondrial DNA and ribosomes.

## Cellular respiration

- Cellular respiration is the process of releasing energy from organic molecules, such as glucose, by breaking them down into simpler molecules, such as carbon dioxide and water.
- Cellular respiration can be divided into two types: aerobic respiration and anaerobic respiration.
- Aerobic respiration is the most efficient and common type of cellular respiration, which requires oxygen as the final electron acceptor in the electron transport chain.
- Anaerobic respiration is a less efficient and less common type of cellular respiration, which occurs in the absence of oxygen or when oxygen is limited.
- Anaerobic respiration uses alternative electron acceptors, such as sulfate, nitrate, or fumarate, in the electron transport chain, or bypasses the electron transport chain altogether by producing fermentation products, such as ethanol or lactic acid.

## Relationship of carbohydrate metabolism to other compounds

- Carbohydrate metabolism is the set of biochemical reactions that involve the synthesis and degradation of carbohydrates, such as glucose, glycogen, starch, and cellulose.
- Carbohydrate metabolism is closely related to the metabolism of other compounds, such as lipids, proteins, and nucleic acids, because they can be interconverted by various pathways.
- For example, glucose can be converted into fatty acids and glycerol, which are the building blocks of lipids, or into amino acids, which are the building blocks of proteins, or into ribose, which is the building block of nucleic acids.
- Conversely, lipids, proteins, and nucleic acids can be converted into glucose or other intermediates of carbohydrate metabolism, such as pyruvate, acetyl-CoA, or oxaloacetate, by various pathways.
- The interconversion of different compounds allows cells to regulate their energy balance and to adapt to different environmental conditions.

## Glycolysis

- Glycolysis is the first stage of glucose oxidation, which occurs in the cytoplasm of all cells .
- Glycolysis is the process of breaking down one molecule of glucose (a six-carbon sugar) into two molecules of pyruvate (a three-carbon compound) by a series of 10 enzymatic reactions .
- Glycolysis can be divided into two phases: the preparatory phase and the payoff phase .
- The preparatory phase consumes two molecules of ATP to phosphorylate glucose and split it into two molecules of glyceraldehyde-3-phosphate (G3P)[

## Structure of mitochondria

- Mitochondria are small, rod-shaped organelles found in both plant and animal cells that are involved in energy production through cellular respiration .
- Mitochondria have a double membrane structure, consisting of an outer membrane and an inner membrane . The outer membrane is smooth and surrounds the organelle, while the inner membrane is folded into projections called cristae .
- The space between the outer and inner membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix . The matrix contains enzymes, DNA, ribosomes, and other molecules that are essential for mitochondrial function .
- The inner membrane also contains proteins called electron transport

chains, which are involved in the generation of ATP, the main energy currency of the cell . The electron transport chains use the energy from the oxidation of organic molecules, such as glucose, to pump protons across the inner membrane, creating a proton gradient .

- The protons then flow back into the matrix through a protein complex called ATP synthase, which uses the proton motive force to synthesize ATP from ADP and phosphate . This process is called oxidative phosphorylation, and it is the final stage of cellular respiration .
- The other stages of cellular respiration, namely glycolysis, the formation of acetyl co-A, and the Krebs cycle, occur in the cytoplasm or the matrix of the mitochondria, depending on the type of cell and the availability of oxygen . These stages produce the organic molecules and the electrons that are used by the electron transport chains to generate ATP .
- The amount of ATP produced by mitochondria depends on several factors, such as the type of substrate, the oxygen level, the temperature, the pH, and the presence of inhibitors or uncouplers . Mitochondria can also regulate their own activity and number by various mechanisms, such as fusion, fission, biogenesis, and autophagy.
- Mitochondria are essential for the survival and function of most eukaryotic cells, as they provide the energy needed for various cellular processes, such as growth, differentiation, signaling, and apoptosis . Mitochondria also have other roles, such as calcium homeostasis, iron metabolism, reactive oxygen species production, and innate immunity .
- Mitochondria are thought to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells in a process called endosymbiosis . This is supported by the fact that mitochondria have their own DNA, which is circular and similar to bacterial DNA, and their own ribosomes, which are similar to bacterial ribosomes .
- Mitochondrial DNA is inherited maternally in most organisms, and it can be used to trace the evolutionary history and the genetic diversity of populations . Mitochondrial DNA can also mutate and cause various diseases, such as mitochondrial myopathies, neurodegenerative disorders, diabetes, and cancer .

Hello, I am Sydney, your AI assistant. I can help you with your topic of cellular respiration. Here are some notes that you can use for your study material.

### **Cellular respiration**

- Cellular respiration is the process by which biological fuels are oxidised in the presence of an inorganic electron acceptor such as oxygen to produce large amounts of energy, to drive the bulk production of adenosine triphosphate (ATP).
- ATP is the main energy currency of the cell, and it is used to power various cellular processes such as biosynthesis, transport, and movement.
- Cellular respiration can be divided into four stages: glycolysis, pyruvate

oxidation, the citric acid or Krebs cycle, and oxidative phosphorylation.

### **Structure of mitochondria**

- Mitochondria are the organelles where most of the cellular respiration takes place in eukaryotic cells.
- Mitochondria have a double membrane structure, with an outer membrane and an inner membrane.
- The outer membrane is smooth and permeable to most molecules, while the inner membrane is folded into cristae that increase the surface area for the electron transport chain and ATP synthase.
- The space between the two membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix.
- The matrix contains enzymes for the Krebs cycle, mitochondrial DNA, and ribosomes.

### **Glycolysis**

- Glycolysis is the first stage of cellular respiration, and it takes place in the cytoplasm of the cell.
- Glycolysis is the breakdown of glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis involves ten enzymatic reactions, and it can be divided into two phases: the energy investment phase and the energy payoff phase.
- In the energy investment phase, two molecules of ATP are used to phosphorylate glucose and its intermediates, making them more reactive.
- In the energy payoff phase, four molecules of ATP and two molecules of NADH (a reduced form of nicotinamide adenine dinucleotide, an electron carrier) are produced per glucose molecule.
- The net gain of glycolysis is two molecules of ATP and two molecules of NADH per glucose molecule.
- Glycolysis can occur in the presence or absence of oxygen. If oxygen is available, pyruvate will enter the mitochondria for further oxidation. If oxygen is not available, pyruvate will undergo fermentation to regenerate NAD<sup>+</sup> for glycolysis.

### **Pyruvate oxidation**

- Pyruvate oxidation is the second stage of cellular respiration, and it takes place in the matrix of the mitochondria.
- Pyruvate oxidation is the conversion of pyruvate into acetyl coenzyme A (acetyl-CoA), a two-carbon molecule that can enter the Krebs cycle.
- Pyruvate oxidation involves three steps: the decarboxylation of pyruvate, the oxidation of the remaining two-carbon fragment, and the transfer of the acetyl group to coenzyme A.
- Pyruvate oxidation produces one molecule of CO<sub>2</sub>, one molecule of NADH, and one molecule of acetyl-CoA per pyruvate molecule.

- Pyruvate oxidation requires oxygen as the final electron acceptor for NADH.

### **Krebs cycle**

- The Krebs cycle, also known as the citric acid cycle, is the third stage of cellular respiration, and it takes place in the matrix of the mitochondria.
- The Krebs cycle is a series of eight reactions that oxidise acetyl-CoA into two molecules of CO<sub>2</sub>, while generating ATP, NADH, and FADH<sub>2</sub> (a reduced form of flavin adenine dinucleotide, another electron carrier).
- The Krebs cycle begins with the condensation of acetyl-CoA with oxaloacetate, a four-carbon molecule, to form citrate, a six-carbon molecule.
- The Krebs cycle then involves a series of decarboxylations, dehydrogenations, and rearrangements, resulting in the regeneration of oxaloacetate and the release of two molecules of CO

### **Relationship of carbohydrate metabolism to other compounds**

- Carbohydrate metabolism is the process of breaking down and synthesizing carbohydrates, which are organic molecules composed of carbon, hydrogen, and oxygen atoms.
- Carbohydrates can be classified into simple sugars (monosaccharides and disaccharides) and complex sugars (polysaccharides), such as starch, glycogen, and cellulose.
- Carbohydrate metabolism is connected to the metabolism of other compounds, such as proteins and lipids, through common intermediates and pathways .
- The main pathway of carbohydrate metabolism is glycolysis, which converts glucose into pyruvate, generating ATP and NADH.
- Pyruvate can then enter the mitochondria and be converted into acetyl coenzyme A (acetyl CoA), which is the entry point for the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle).
- The citric acid cycle is a series of reactions that produces more ATP, NADH, FADH<sub>2</sub>, and carbon dioxide.
- The NADH and FADH<sub>2</sub> then donate their electrons to the electron transport chain, which is a series of protein complexes embedded in the inner mitochondrial membrane.
- The electron transport chain transfers the electrons along a gradient of proton pumps, creating a proton motive force that drives the synthesis of ATP by the enzyme ATP synthase.
- This process is called oxidative phosphorylation, because it involves the oxidation of NADH and FADH<sub>2</sub> and the phosphorylation of ADP to ATP.
- The metabolism of other compounds can also feed into or out of carbohydrate metabolism through various connections .
- For example, the simple sugars galactose and fructose can be converted into glucose or intermediates of glycolysis, such as glyceraldehyde-3-

phosphate or dihydroxyacetone phosphate .

- Glycogen, which is a polysaccharide that stores glucose in animals, can be broken down into glucose-1-phosphate and then converted into glucose-6-phosphate, which is the first intermediate of glycolysis .
- Pentose, which is a five-carbon sugar, can be derived from the pentose phosphate pathway, which is a branch of glycolysis that generates NADPH and ribose-5-phosphate .
- Ribose-5-phosphate is the precursor for the synthesis of nucleotides, such as ATP, GTP, CTP, and UTP .
- The amino acids from proteins can also connect with carbohydrate metabolism through pyruvate, acetyl CoA, and components of the citric acid cycle .
- Depending on their structure, amino acids can be classified into glucogenic, ketogenic, or both .
- Glucogenic amino acids can be converted into pyruvate or intermediates of the citric acid cycle, such as oxaloacetate, fumarate, succinyl CoA, or alpha-ketoglutarate .
- These intermediates can then be used to synthesize glucose through a process called gluconeogenesis, which is the reverse of glycolysis .
- Ketogenic amino acids can be converted into acetyl CoA or acetoacetyl CoA, which are the precursors for the synthesis of ketone bodies, such as acetoacetate, beta-hydroxybutyrate, and acetone .
- Ketone bodies are an alternative fuel source for the brain and other tissues when glucose is scarce, such as during fasting or starvation .
- Some amino acids, such as leucine, lysine, isoleucine, phenylalanine, tyrosine, and tryptophan, are both glucogenic and ketogenic, meaning they can be converted into either pyruvate, acetyl CoA, or citric acid cycle intermediates<sup>[1]</sup>

## Glycolysis

- Glycolysis is a metabolic pathway that converts glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis does not require oxygen and can occur in both aerobic and anaerobic conditions. In anaerobic conditions, pyruvate is further converted into lactic acid or ethanol.
- Glycolysis is an ancient metabolic pathway that is found in most living organisms. It is the first stage of cellular respiration, which produces energy in the form of ATP and NADH.
- Glycolysis consists of ten enzyme-catalyzed reactions that can be divided into two phases: the preparatory phase and the payoff phase.
- In the preparatory phase, two molecules of ATP are used to phosphorylate glucose and split it into two molecules of glyceraldehyde-3-phosphate (G3P).
- In the payoff phase, each G3P molecule is oxidized and phosphorylated, producing two molecules of NADH and four molecules of ATP. The final

product is two molecules of pyruvate.

- The net gain of glycolysis is two molecules of ATP and two molecules of NADH per molecule of glucose.
- The following table summarizes the steps of glycolysis, the enzymes involved, and the products formed.

| Step | Enzyme                                   | Reaction                                                                                              | Product                                                   |
|------|------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1    | Hexokinase                               | Glucose + ATP<br>-> Glucose-6-phosphate + ADP                                                         | Glucose-6-phosphate                                       |
| 2    | Phosphoglucose isomerase                 | Glucose-6-phosphate -> Fructose-6-phosphate                                                           | Fructose-6-phosphate                                      |
| 3    | Phosphofructokinase                      | Fructose-6-phosphate + ATP -> Fructose-1,6-bisphosphate + ADP                                         | Fructose-1,6-bisphosphate                                 |
| 4    | Aldolase                                 | Fructose-1,6-bisphosphate -> Dihydroxyacetone phosphate + Glyceraldehyde-3-phosphate                  | Dihydroxyacetone phosphate and Glyceraldehyde-3-phosphate |
| 5    | Triose phosphate isomerase               | Dihydroxyacetone phosphate -> Glyceraldehyde-3-phosphate                                              | Glyceraldehyde-3-phosphate                                |
| 6    | Glyceraldehyde-3-phosphate dehydrogenase | Glyceraldehyde-3-phosphate + NAD <sup>+</sup> + Pi -> 1,3-Bisphosphoglycerate + NADH + H <sup>+</sup> | 1,3-Bisphosphoglycerate and NADH                          |
| 7    | Phosphoglycerate kinase                  | 1,3-Bisphosphoglycerate + ADP -> 3-Phosphoglycerate + ATP                                             | 3-Phosphoglycerate and ATP                                |
| 8    | Phosphoglycerate mutase                  | 3-Phosphoglycerate -> 2-Phosphoglycerate                                                              | 2-Phosphoglycerate                                        |

| Step | Enzyme          | Reaction                                                        | Product             |
|------|-----------------|-----------------------------------------------------------------|---------------------|
| 9    | Enolase         | 2-Phosphoglycerate<br>-> Phosphoenolpyruvate + H <sub>2</sub> O | Phosphoenolpyruvate |
| 10   | Pyruvate kinase | Phosphoenolpyruvate + ADP -> Pyruvate + ATP                     | Pyruvate and ATP    |

### Formation of Acetyl Co-A

- Acetyl Co-A (acetyl coenzyme A) is a molecule that participates in many biochemical reactions in protein, carbohydrate and lipid metabolism.
- Its main function is to deliver the acetyl group to the citric acid cycle (Krebs cycle) to be oxidized for energy production.
- Acetyl Co-A formation occurs inside or outside the cell mitochondria. As a metabolite (a substance necessary for metabolism), acetyl Co-A must be freely available.
- It can be produced via the catabolism (breakdown) of carbohydrates (glucose) and lipids (fatty acids).
- Acetyl Co-A formation most commonly occurs during glucose catabolism. After carbohydrates have been broken down by digestive enzymes, the first stage of cellular glucose metabolism or glycolysis can begin. Glycolysis is the breaking down of glucose molecules.
- Glycolysis produces two molecules of pyruvate (a three-carbon compound) per molecule of glucose. Pyruvate then enters the mitochondria, where it is converted into acetyl Co-A by a multi-enzyme complex called pyruvate dehydrogenase.
- The conversion of pyruvate to acetyl Co-A involves the removal of a carboxyl group as carbon dioxide and the transfer of an acetyl group to coenzyme A. This reaction also produces one molecule of NADH (a high-energy electron carrier) per pyruvate.
- The overall formation reaction of acetyl Co-A may be represented as:  

$$\text{pyruvic acid} + \text{CoA} + \text{NAD}^+ \rightarrow \text{acetyl Co-A} + \text{NADH} + \text{H}^+ + \text{CO}_2$$
- This reaction may be called the oxidative decarboxylation of pyruvic acid to acetyl Co-A.
- Acetyl Co-A then enters the citric acid cycle, where it combines with a four-carbon molecule and goes through a cycle of reactions, ultimately regenerating the four-carbon starting molecule.



- The citric acid cycle produces ATP (or, in some cases, GTP), NADH, and FADH<sub>2</sub> (another high-energy electron carrier), and releases carbon dioxide.
- The NADH and FADH<sub>2</sub> molecules then donate their electrons to the electron transport chain, where a series of redox reactions generate a proton gradient across the inner mitochondrial membrane.
- The proton gradient drives the synthesis of ATP by a membrane-bound enzyme called ATP synthase. This process is called oxidative phosphorylation.
- The formation of acetyl Co-A is a key step in cellular respiration, as it links glycolysis to the citric acid cycle and the electron transport chain.
- The formation of acetyl Co-A is also influenced by various factors, such as the availability of oxygen, glucose, fatty acids, and other substrates, as well as the hormonal and enzymatic regulation of the metabolic pathways.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here are some notes on the Krebs cycle:

### Krebs cycle

- The Krebs cycle, also known as the citric acid cycle, is a series of chemical reactions that occur in the mitochondria of aerobic organisms .
- The Krebs cycle is part of the cellular respiration process, which converts glucose and oxygen into carbon dioxide, water, and energy (ATP) .
- The Krebs cycle requires pyruvate, which is the product of glycolysis, the first stage of cellular respiration .
- Pyruvate is converted into acetyl-CoA, a two-carbon molecule, by the enzyme pyruvate dehydrogenase complex .
- Acetyl-CoA enters the Krebs cycle by combining with oxaloacetate, a four-carbon molecule, to form citrate, a six-carbon molecule .
- Citrate undergoes a series of eight steps, involving various enzymes and coenzymes, to regenerate oxaloacetate and produce two molecules of carbon dioxide, three molecules of NADH, one molecule of FADH<sub>2</sub>, and one molecule of ATP per cycle .
- NADH and FADH<sub>2</sub> are electron carriers that transfer their electrons to the electron transport chain, the final stage of cellular respiration, where most of the ATP is generated by oxidative phosphorylation .
- The Krebs cycle also provides precursors for the synthesis of certain amino acids, as well as other biomolecules .
- The Krebs cycle is regulated by feedback inhibition, where the products of the cycle inhibit the enzymes that catalyze the cycle .
- The Krebs cycle is one of the oldest metabolic pathways, and is conserved across many different organisms .

: [https://simple.wikipedia.org/wiki/Krebs\\_cycle](https://simple.wikipedia.org/wiki/Krebs_cycle) [https://chem.libretexts.org/Bookshelves/Biological\\_](https://chem.libretexts.org/Bookshelves/Biological_)

[https://en.wikipedia.org/wiki/Citric\\_acid\\_cycle](https://en.wikipedia.org/wiki/Citric_acid_cycle)  
<https://www.khanacademy.org/science/biology/cellular-respiration-and-fermentation/pyruvate-oxidation-and-the-citric-acid-cycle/a/the-citric-acid-cycle>  
<https://biologydictionary.net/krebs-cycle/>

## Electron Transport System and Oxidative Phosphorylation

- Electron transport system (ETS) or electron transport chain (ETC) is a series of proteins and electron carriers embedded in the inner mitochondrial membrane that transfer electrons from reduced molecules such as NADH and FADH<sub>2</sub> to oxygen, the final electron acceptor .
- Oxidative phosphorylation (OXPHOS) is the process of synthesizing ATP from ADP and inorganic phosphate (Pi) using the energy released by the ETS .
- The ETS consists of four protein complexes (I, II, III, and IV) and two mobile electron carriers (ubiquinone and cytochrome c) that shuttle electrons between the complexes .
- Complex I (NADH dehydrogenase) receives electrons from NADH and transfers them to ubiquinone, while pumping four protons (H<sup>+</sup>) across the membrane .
- Complex II (succinate dehydrogenase) receives electrons from FADH<sub>2</sub> and transfers them to ubiquinone, without pumping protons .
- Ubiquinone (also called coenzyme Q or Q) is a lipid-soluble molecule that can diffuse within the membrane and carry electrons from complex I or II to complex III .
- Complex III (cytochrome bc<sub>1</sub> complex) receives electrons from ubiquinone and transfers them to cytochrome c, while pumping four protons across the membrane .
- Cytochrome c is a water-soluble protein that can move along the intermembrane space and carry electrons from complex III to complex IV .
- Complex IV (cytochrome c oxidase) receives electrons from cytochrome c and transfers them to oxygen, the final electron acceptor, while pumping two protons across the membrane .
- Oxygen is reduced to water by accepting four electrons and four protons at complex IV .
- The ETS creates an electrochemical gradient across the inner mitochondrial membrane, with a higher concentration of protons and a lower pH in the intermembrane space than in the matrix .
- The electrochemical gradient provides the potential energy for ATP synthesis by chemiosmosis, the movement of protons down their gradient through a protein channel called ATP synthase .
- ATP synthase consists of two subunits: F<sub>0</sub> and F<sub>1</sub>. F<sub>0</sub> is embedded in the membrane and acts as a proton channel. F<sub>1</sub> is exposed to the matrix and acts as an enzyme that catalyzes the phosphorylation of ADP to ATP .
- As protons flow through F<sub>0</sub>, they cause a rotation of a central stalk that

connects F0 and F1. The rotation of the stalk induces conformational changes in F1 that allow it to bind ADP and Pi and release ATP .

- The ETS and OXPHOS are coupled by the proton-motive force, the combined effect of the proton concentration gradient and the electric potential difference across the membrane .
- The ETS and OXPHOS are the final stages of cellular respiration, the process of breaking down organic molecules such as glucose to release energy and produce ATP .
- Cellular respiration consists of four main steps: glycolysis, pyruvate oxidation, citric acid cycle (or Krebs cycle), and ETS and OXPHOS .
- Glycolysis is the breakdown of glucose into two molecules of pyruvate, with the net production of two ATP and two NADH molecules per glucose molecule .
- Pyruvate oxidation is the conversion of pyruvate into acetyl-CoA, with the production of one CO<sub>2</sub> and one NADH molecule per pyruvate molecule

## ATP

- ATP stands for adenosine triphosphate, an organic compound that consists of a nitrogenous base (adenine), a sugar (ribose), and three phosphate groups.
- ATP is the main energy currency of the cell, as it can store and release energy quickly by breaking and forming the bonds between the phosphate groups.
- ATP is produced by cellular respiration, a process that oxidizes glucose and other organic molecules to generate ATP and release carbon dioxide and water as waste products.
- Cellular respiration can be divided into three main stages: glycolysis, the citric acid cycle (or Krebs cycle), and oxidative phosphorylation .
  - Glycolysis is the first stage of cellular respiration, where glucose is split into two molecules of pyruvate, producing a net of 2 ATP and 2 NADH (a coenzyme that carries electrons) per glucose molecule .
  - The citric acid cycle is the second stage of cellular respiration, where pyruvate is converted into acetyl-CoA, which then enters a cycle of reactions that produces 2 ATP, 6 NADH, and 2 FADH<sub>2</sub> (another coenzyme that carries electrons) per glucose molecule .
  - Oxidative phosphorylation is the third and final stage of cellular respiration, where the electrons from NADH and FADH<sub>2</sub> are transferred to a series of proteins called the electron transport chain, which pumps protons across the inner membrane of the mitochondria, creating a gradient that drives the synthesis of ATP by a protein complex called ATP synthase. This stage produces about 26-28 ATP per glucose molecule .
- The structure of the mitochondria, the organelles where most of cellular respiration takes place, is adapted to facilitate the process. The mitochondria have a double membrane, with the inner membrane folded into cristae

that increase the surface area for the electron transport chain and ATP synthase. The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the matrix, where the citric acid cycle occurs.

- The rate of cellular respiration is affected by various factors, such as the availability of glucose and oxygen, the temperature, the pH, and the presence of inhibitors or activators of the enzymes involved in the process.